Exponential growth and decay

1 For each of the following functions:

i Specify if the function shows exponential growth or exponential decay.

ii Find the rate of growth or decay.

a $y = 0.68\,(1.6)^x$

b $y = 0.98\,(0.2)^x$

2 Tina currently follows 174 people on Twitter. The number of people she follows is growing at a rate of 5% per month. How many people will she be following in 6 months?

3 A saving account initially has $\$26\,030$. The account increases by 1.1% annually.
State the function, $f(t)$, for the amount in the savings account after t years.

4 The population of a country can be modeled by $P(t) = 10\,637\,(1.0008)^{5t}$, where $P(t)$ is the population after t years.
Rewrite $P(t)$ in the form $P(t) = 10\,637\,(\square)^t$.

5 The local rat population numbers approximately 750 and is increasing at a rate of 3% per year.

a Write a function, y, to represent the population after t years.

b Find the size of the population after 5 years, rounding to the nearest whole number.

6 Justin purchased a piece of sports memorabilia for $\$2900$, and it is expected to increase in value by 9% per year.

a Write a function y to represent the value of the piece of sports memorabilia after v years.

b Find the value of the piece of sports memorabilia after 8 years.

7 Due to favorable market conditions, a company's annual costs are expected to decrease by 5% per year for the next few years. Their annual costs for the current financial year are $\$322\,000$.

a Write a function, y, to represent the company's annual costs in m years.

b Find the company's annual costs in 5 years.

8 The number of martial arts clubs worldwide [redacted]pproximately 2310 in 1988 and has been decreasing at a rate of 2.6% per year since.

a Write a function, y, to represent the number of martial arts clubs n years after 1988.

b Find the number of martial arts clubs in 2009. Round your answer to the nearest whole number.

9 Ryan has been earning a decent income from playing online poker. He started out with $\$200$ and has been able to triple his money every year.

a Write a function to represent the amount of money he had, y, after n years.

b Find the amount of money he had after 5 years of playing online poker.

10 A ball dropped from a height of 21 meters will bounce back off the ground to 50% of the height of the previous bounce (or the height from which it is dropped when considering the first bounce).

a Write a function, y, to represent the height of the nth bounce.

b Find the height of the fifth bounce. Round your answer to two decimal places.

11 A sample contains 300 grams of carbon-11, which has a half-life of 20 minutes.

a Write a function, A, to represent the amount of carbon-11 remaining in the sample after t minutes.

b Find how much of the isotope would be left after 3 hours. Round your answer to two decimal places.

12 Determine whether or not the following scenarios involve a limiting value:

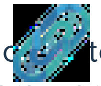
a The amount of radiation in the air after a nuclear disaster if the radiation decreases by half every 100 years.

b The number of employees in a company over the next 10 years.

c Saving $\$100$ each week until the savings target is reached.

d The value of a piece of machinery if it decreases by 30% each year.

13 The population of a town is expected to double in the next 20 years. At what rate, r, will it grow each year, assuming growth is constant? Give your answer as a percentage correct to two decimal places.
Let the current population be k if necessary.

14 A scientist observed that a certain species oteria grew by 13% each hour. How many bacteria will be present after 17 hours if there were 79 initially? Round your answer to the nearest whole number.

15 A new car purchased for $\$38\,200$ depreciates at a rate, r, each year.

a Use the table of values to determine the value of r.

Years passed (n)	0	1	2
Value of car(A)	$38\,200$	$37\,818$	$37\,439.82$

b Determine the rule for A, the value of the car, n years after it is purchased.

c Assuming the rate of depreciation remains constant, how much can the car be sold for after 6 years?

d A new motorbike purchased for the same amount depreciates according to the model $V = 38\,200\,(0.97^n)$. Which vehicle depreciates more rapidly?

16 The population of Joanne's home town is $27\,400$ and has a growth rate of 6% each year. What is the estimated population after 7 years?

17 The population of some native bees is declining at a rate of 8% per year. If there are $10\,000$ bees in a hive now, how many will there be in 5 years time?

18 The population of cows at a free range farm has a growth rate of 17%. If the farm originally has 140 cows, how many will there be after 4 years?

19 An online business that sells phones online [redacted]een growing very quickly over the last year. They have asked you to model their growth over the next three quarters, and further into the future.

a Looking at the quarterly data, you start by assuming the number of customers changed in the same way each quarter over the first year.
Complete the table under this assumption to predict the number of phones sold over the next three quarters.

Quarter(N)	1	2	3	4	5	6	7
Online Customers (C)	480	2880	17280	$103\,680$			

b Find an equation for the number of sales, C, in terms of the number of quarters that have passed, N.

c Use the formula to predict how many sales they will have 2.75 years after they started the business.

d The company tells you that they do not expect to sell the predicted number of phones in part (c) in a single quarter 2.75 years from now. What should you tell the company?